Every rational proportion of good vertices: β-Kirchhoff Šoltés graphs for all $\beta < 1$

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Abstract. A vertex v of a connected graph G is good if $G - v$ is connected and $\mathrm{Kf}(G - v) = \mathrm{Kf}(G)$, where Kf is the Kirchhoff index. Gottwald, Majstorović Ergotić and Došlić asked, for a fixed rational $\beta \in (0, 1]$, for an infinite series of graphs in which at least a proportion β of the vertices are good; their best proportions tend to $2/3$. We prove that every rational $\beta \in (0, 1)$ is the exact proportion of good vertices of infinitely many connected graphs. This settles their Problem 4 for every $\beta < 1$ and their Problem 5 for every $\gamma \in (0, 1]$, and it answers their Problem 17: the exact values are the rationals in $(0, 1]$ and the limiting values are all real numbers in $(0, 1]$. The graphs are a clique K_{ab+1} with one extra vertex (a hub) attached to each of a disjoint blocks of b clique vertices, and a small rooted graph W glued at the clique vertex outside the blocks. The good vertices are exactly the block vertices. Their goodness equation is linear in the transmission of W, and all its denominators are explicit products of small numbers, so it can be solved exactly. Explicit examples: a graph on 1772 vertices with proportion $297/443 > 2/3$ and a graph on 2277 vertices with proportion exactly $2/3$, both checked by exact arithmetic on the actual Laplacian. The case $\beta = 1$ (the Kirchhoff Šoltés problem itself) is not addressed.

The question

K. K. Gottwald, S. Majstorović Ergotić, T. Došlić, *Šoltés problem for the Kirchhoff index of a graph*, Applied Mathematics and Computation 510 (2026) 129694 [1] (arXiv:2609.13751). With $K(G) = \{v \in V(G) : \mathrm{Kf}(G) = \mathrm{Kf}(G - v)\}$, they call G a β-Kirchhoff Šoltés graph if $|K(G)| \geq \beta |V(G)|$, and they ask (journal pp. 3 and 11):

> *"Problem 4. For a fixed rational number $\beta \in (0, 1]$ construct an infinite series of β-Kirchhoff Šoltés graphs."*
> *"Problem 5. Given a fixed $\gamma \in (0, 1]$, construct an infinite series of graphs where the proportion of good vertices increases and asymptotically approaches γ as the order of the graph grows."*
> *"Problem 17. Which (rational or irrational) numbers $\gamma \in (0, 1]$ appear either as exact or as limiting values of proportions of good vertices with respect to the Kirchhoff Šoltés problem for simple undirected graphs?"*

A vertex whose removal disconnects the graph is not good. The largest limiting proportion in [1] is $2/3$, from splices of $K_4 - e$. The earlier stage of the present work [2] proved that C_5 is the only connected graph with at most 10 vertices and proportion at least $2/3$, and that splices of bounded branches cannot exceed $2/3$ (for branches with at most 9 vertices, by exact computation). It left $\beta \geq 2/3$ open.

Theorem 1. For every rational $\beta \in (0, 1)$ there are connected graphs G of arbitrarily large order with $|K(G)| = \beta\,|V(G)|$.

Corollary 2. **(1)** Problem 4 holds for every rational $\beta \in (0, 1)$.

(2) Problem 5 holds for every real $\gamma \in (0, 1]$: there are graphs of strictly increasing orders whose proportions of good vertices increase strictly and tend to γ.

(3) Problem 17: the exact proportions are exactly the rational numbers in $(0, 1]$, and every real number in $(0, 1]$ is a limiting proportion.

Proof. (1) follows from Theorem 1, since orders are unbounded. For (2) choose rationals $\beta_1 < \beta_2 < \dots$ in $(0, 1)$ tending to γ, and inductively a graph G_i with proportion exactly β_i and $|V(G_i)| > |V(G_{i-1})|$. (3) follows, with 1 attained by C_5 [1]; an exact proportion is rational by definition. □

The case $\beta = 1$ of Problem 4 is the Kirchhoff Šoltés problem (Problem 3 of [1]), which we do not address [3]: all graphs below have cut vertices, so their proportion is below 1.

1 Notation and the splice lemma

Graphs are finite, simple and connected. $r_G(u, w)$ is the effective resistance between u and w when each edge is a unit resistor, $\mathrm{Rt}_G(u) = \sum_w r_G(u, w)$ is the transmission, and $\mathrm{Kf}(G) = \sum_{\{u,w\}} r_G(u, w)$ [4] [5]. If L is the Laplacian of G and $L\varphi = e_s - e_t$, then $r_G(s, t) = \varphi(s) - \varphi(t)$. We write $\Delta_G(v) = \mathrm{Kf}(G) - \mathrm{Kf}(G - v)$ when $G - v$ is connected, so v is good iff $\Delta_G(v) = 0$.

A rooted graph (Z, x) is a graph with a root x. Put $m_Z = |V(Z)| - 1$ and $T_Z = \mathrm{Rt}_Z(x)$. For $z \neq x$ with $Z - z$ connected, let $\delta_Z(z) = \mathrm{Kf}(Z) - \mathrm{Kf}(Z - z)$ and $\rho_Z(z) = \mathrm{Rt}_{Z-z}(x) - \mathrm{Rt}_Z(x)$. The splice $Z \oplus Z'$ identifies the roots; m and T are additive under splicing.

Lemma 3 (splice lemma). (Lemma 1 of [2].) Let $G = Z \oplus R$, and let $z \neq x$ be a vertex of Z. If $Z - z$ is connected, then $\Delta_G(z) = \delta_Z(z) - m_R\,\rho_Z(z) + T_R$. If $Z - z$ is disconnected, then z is a cut vertex of G, so it is not good.

Proof. Resistances inside Z and inside R are unchanged in G, and $r_G(u, w) = r_Z(u, x) + r_R(x, w)$ across x. Hence $\mathrm{Kf}(Z \oplus R) = \mathrm{Kf}(Z) + \mathrm{Kf}(R) + m_R T_Z + m_Z T_R$. Subtract the same identity for $(Z - z) \oplus R$, in which $m_{Z-z} = m_Z - 1$. □

2 The clique with hubs

Let U be a set of $N \geq 2$ vertices, split into blocks $B_1, \dots, B_a$ of sizes $b_i \geq 1$ and a set C of hubless vertices. The graph H is the clique on U together with hubs $h_1, \dots, h_a$, where h_i is adjacent exactly to the vertices of B_i. Put $x_i = 1/b_i$, $Y = \sum_i x_i$ and $a' = a + |C|$. This class is closed under the deletion of a vertex of U, and under the deletion of a hub (the vertices of its block become hubless). It is what makes everything below explicit: eliminating the hubs from the Laplacian leaves a matrix on U with the eigenvalues 0, N and $N + 1$ only, whatever the block sizes.

Lemma 4. Let f be a function on $V(H)$ with $\sum f = 0$, and let $F_i = f(h_i) +$

$\sum_{u\in B_i} f(u)$. Then $L_H\varphi = f$ for

$$\varphi(h_i) = \frac{F_i}{b_i N} + \frac{f(h_i)}{b_i}\,, \quad \varphi(u) = \frac{f(u)+\varphi(h_i)}{N+1} \ (u\in B_i), \quad \varphi(c) = \frac{f(c)}{N} \ (c\in C).$$

Consequently, for $u \in B_i$, $u' \in B_j$, and hubless $c \neq c'$:

$$r(u,u') = \frac{2}{N+1} \quad (i=j), \qquad r(u,u') = \frac{2+(x_i+x_j)/N}{N+1} \quad (i\neq j),$$
$$r(h_i,u) = x_i + \frac{1-x_i}{N+1}\,, \qquad r(h_i,u') = x_i\,\frac{N+1}{N} + \frac{1+x_j/N}{N+1} \quad (i\neq j),$$
$$r(h_i,h_j) = (x_i+x_j)\,\frac{N+1}{N} \quad (i\neq j), \qquad r(c,c') = \frac{2}{N}\,,$$
$$r(c,u) = \frac{1}{N} + \frac{1+x_i/N}{N+1}\,, \qquad r(c,h_i) = \frac{1}{N} + x_i\,\frac{N+1}{N}\,.$$

Proof. From the definition, $\sum_{u\in B_i}\varphi(u) = (F_i - f(h_i) + b_i\varphi(h_i))/(N+1) = F_i/N$, so $\sum_U \varphi = \frac{1}{N}\left(\sum_i F_i + \sum_C f\right) = 0$. Hence $L_H\varphi(u) = (N+1)\varphi(u) - \varphi(h_i) = f(u)$ on B_i, $L_H\varphi(c) = N\varphi(c) = f(c)$ on C, and $L_H\varphi(h_i) = b_i\varphi(h_i) - F_i/N = f(h_i)$. The resistances follow by taking $f = e_s - e_t$ and $r(s,t) = \varphi(s) - \varphi(t)$. □

A hubless vertex behaves in these formulas like a block of size 1 without its hub: $r(c,u)$ and $r(c,c')$ are the second formula of the list with $x = 1$, and $r(c,h_i)$ is the fourth with $x_j = 1$.

Lemma 5.

$$\mathrm{Kf}(H) = Y\,\frac{(N+1)(N+a-1)}{N} + (N+a-2) + \frac{(1+a')(1-a)}{N+1} + \frac{a(a'-2)}{N}\,,$$

and for a hubless vertex c,

$$\mathrm{Rt}_H(c) = 2 + Y + \frac{Y+a+a'-2}{N} - \frac{1+a'}{N+1}\,.$$

Proof. Call parts the blocks and the singletons of C, with $x_P = 1/|P|$. We use $\sum_{P<P'}|P||P'|(x_P+x_{P'}) = (a'-1)N$ and sum the resistances of Lemma 4. Pairs in U give $(N(N-1)+a'-1)/(N+1)$. A hub h_i and U give $1+\frac{N-1}{N+1}+x_i(N+1)-\frac{N+1}{N}+\frac{a'-1}{N(N+1)}$. Pairs of hubs give $(a-1)Y(N+1)/N$. Adding and using $N(N-1)+a'-1+a(N-a') = (N+1)(N+a-2)+(1+a')(1-a)$ gives $\mathrm{Kf}(H)$. For $\mathrm{Rt}_H(c)$, the vertices of U give $(2(N-1)+(N+a'-2)/N)/(N+1)$ and the hubs give $a/N + Y(N+1)/N$; partial fractions in N give the formula. □

From now on $H = H_{a,b}$, for $a \geq 2$, $b \geq 2$: all a blocks have size b, and $C = \{w\}$. So $N = ab+1$, $a' = a+1$, $Y = a/b$, and $|V(H)| = ab+a+1$. Put $B = b(b-1)$. Removing a block vertex v gives the parameters $N-1$, a, $a+1$, $Y+1/B$. Removing a hub h gives N, $a-1$, $a+b$, $Y-1/b$. Let

$$\delta = \mathrm{Kf}(H) - \mathrm{Kf}(H-v), \qquad \rho = \mathrm{Rt}_{H-v}(w) - \mathrm{Rt}_H(w),$$

$$\delta_h = \mathrm{Kf}(H) - \mathrm{Kf}(H-h), \qquad \rho_h = \mathrm{Rt}_{H-h}(w) - \mathrm{Rt}_H(w).$$

Lemma 6. These numbers do not depend on the choice of v or h, and

$$\begin{aligned}
\delta &= -\frac{2a}{B} + \frac{b^4-b^3-2b+1}{b^3(b-1)} + \frac{2(b-1)(b+2)}{b^2(ab+2)} \\
&\quad - \frac{(b+1)(2b^2-2b-1)}{b^3(ab+1)} + \frac{1}{ab^2(b-1)}\,, \\
\rho &= \frac{1}{B} + \frac{2(b-1)}{b(ab+2)} - \frac{b^2-3b-1}{b^2(ab+1)} - \frac{b^2-b-1}{ab^2(b-1)}\,, \\
\delta_h &= \frac{(N+1)(N+2a-2)}{bN} + 1 + \frac{ab-2a-2b}{N+1} - \frac{(a-1)(b-2)}{N}\,, \\
\rho_h &= -\frac{1}{b} + \frac{b-2-1/b}{N} - \frac{b-1}{N+1}\,.
\end{aligned}$$

Moreover $\delta_h \geq a-2$, and $\rho_h < 0$ when $N \geq b$.

Proof. Substitute the parameters in Lemma 5. These are identities between rational functions of a and b; they are checked symbolically, and on actual graphs, by `check_formulas.py` (claims F3, F4). The estimates: the first term of δ_h is at least $N/b \geq a$, the third is at least -2 and the last at least -1. In ρ_h, $(b-2-1/b)/N - (b-1)/(N+1) = \big((b-2-1/b) - N(1+1/b)\big)/(N(N+1)) < 0$ when $N \geq b$. □

So for fixed b and $a \to \infty$: $\delta = -2a/B + O(1)$ and $\rho = 1/B + O(1/a)$. In particular $\delta < 0$: for large a, deleting a block vertex raises the Kirchhoff index, because the resistance from its hub to the whole graph rises. A branch glued at w raises Δ back, and the denominators of δ and ρ are small. That is the whole idea.

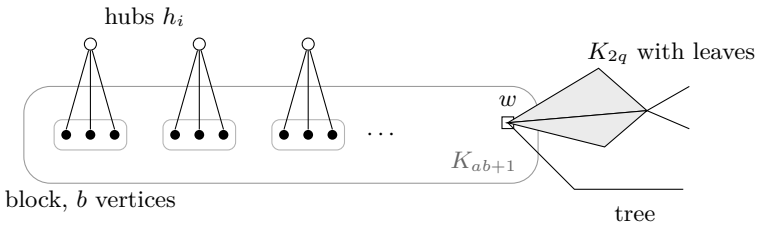


Figure 7 / The graph $G = H_{a,b} \oplus W$. The base is a clique on $ab+1$ vertices. The black block vertices are the good vertices. W is glued at the hubless vertex w: cliques carrying the denominators, cliques that only add vertices, and a tree that fixes the integer part of the transmission.

3 Rooted graphs with prescribed order and transmission

Lemma 8. **(1)** For $c \geq 2$ let K_c be rooted at a vertex. Then $m = c-1$, $T = 2(c-1)/c$, and for each non-root vertex z, $\delta(z) = 1$ and $\rho(z) = -2/(c(c-1))$.

(2) For integers $q \geq 1$, $j \geq 0$, let $Z(q,j)$ be K_{2q} rooted at x, with j pendant leaves attached to one fixed non-root vertex y. Then $m = 2q-1+j$ and

$T = 2 - \frac{1}{q} + j(1 + \frac{1}{q})$. For a clique vertex $z \neq x, y$, with $c = 2q$,

$$\delta(z) = 1 + j + \frac{2j}{c(c-1)}, \qquad \rho(z) = \frac{2(j-1)}{c(c-1)}.$$

(3) For integers $l \geq 1$, $l \leq t \leq l(l+1)/2$, there is a rooted tree with l non-root vertices and $T = t$.

(4) If z is a pendant vertex of Z, $z \neq x$, then $\delta_Z(z) = \mathrm{Rt}_Z(z) > 0$ and $\rho_Z(z) = -r_Z(x, z) < 0$.

Proof. (1) $r = 2/c$ in K_c and $\mathrm{Kf}(K_c) = c - 1$. (2) By Lemma 3's formula for Kf of a splice at y, $\mathrm{Kf}(Z(q, j)) = (c-1) + j^2 + 2j(c-1)/c + j(c-1)$; replace c by $c-1$ for $Z - z$. Also $T = (c-1)2/c + j(1 + 2/c)$. (3) In a tree, T is the sum of the depths. Let k be the largest integer with $k \leq l$ and $k(k+1)/2 + l - k \leq t$. Take a path of k vertices from the root, and $l - k$ further children of the root. If $e = t - k(k+1)/2 - (l-k)$ is positive, then $k < l$ and $e < k$ by maximality of k, and we move one of those children under the path vertex at depth e. (4) is immediate. □

Lemma 9. Let $x \geq 4$ be a multiple of $b(b-1)$, let $a = (x/b)(x^2 - 3x + 1)$, and let

$$Q_0 = b^2(b-1)\,x(x-2), \quad Q_1 = x^2 - 3x + 1,$$

$$Q_2 = x^2 - 2x - 1, \quad Q_3 = x^2 - x - 1, \quad Q_4 = x - 1.$$

Then $ab = xQ_1$, $ab + 1 = Q_4Q_2$, $ab + 2 = (x-2)Q_3$. The numbers $Q_0, \ldots, Q_4$ are pairwise coprime, and the denominators of δ and ρ divide $D = Q_0Q_1Q_2Q_3Q_4$.

Proof. The three identities are expanded directly. Now x is even, since $b(b-1)$ is, so $Q_1, \ldots, Q_4$ are odd, and each prime factor of Q_0 divides $x(x-2)$. Modulo x the numbers $Q_1, \ldots, Q_4$ are $1, -1, -1, -1$, and modulo $x - 2$ they are $-1, -1, 1, 1$; so each is coprime to Q_0. Next $Q_2 - Q_1 = x - 2$, $Q_3 - Q_2 = x$, and $Q_3 - Q_1 = 2(x-1)$; with the odd numbers and the residues above, and $Q_1 \equiv Q_3 \equiv -1 \pmod{x-1}$, this gives $\gcd(Q_1, Q_2) = \gcd(Q_2, Q_3) = \gcd(Q_1, Q_3) = 1$ and $\gcd(Q_1, Q_4) = \gcd(Q_3, Q_4) = 1$. Finally $Q_2 = (x-1)^2 - 2$ and Q_4 is odd, so $\gcd(Q_2, Q_4) = 1$.

The denominators in Lemma 6 are $b(b-1)$, $b^3(b-1)$, $b^2(ab+2) = b^2(x-2)Q_3$, $b^3(ab+1) = b^3Q_4Q_2$, $ab^2(b-1) = b(b-1)xQ_1$, $b(ab+2)$ and $b^2(ab+1)$. Since $b \mid x$, each of them divides D. □

4 Proof of Theorem 1

Let $\beta = p/r$ in lowest terms, $0 < p < r$. Fix an integer $b \geq 2$ with $b(r-p) > p$, and put $B = b(b-1)$ and $\kappa = b(r-p)/p - 1 > 0$. Let s run through the multiples of p, and put $x = Bs$, $a = (b-1)s\,Q_1$ as in Lemma 9, so $p \mid a$. All implied constants below depend only on b and β, and we let $s \to \infty$. Put

$$n = \frac{ab\,r}{p} \in \mathbb{Z}, \qquad m = n - (ab + a + 1) = \kappa a - 1.$$

We shall build a rooted graph (W, w) with $m_W = m$ and $T_W = \tau := m\rho - \delta$. Then $G = H_{a,b} \oplus W$ has n vertices and, by Lemma 3 with $Z = H_{a,b}$, $\Delta_G(v) = \delta - m\rho + \tau = 0$ for each of the ab block vertices v.

Sizes. We have $a \asymp x^3$. Let $E = -\delta$, $\Sigma = \sum_i Q_i = O(x^2)$ and $L = \lceil 2\sqrt{E+1} \rceil = O(x^{3/2})$. By Lemma 6, $E = 2a/B + O(1)$, so $E \asymp x^3$, and $m(\rho - 1/B) = O(1)$. For s large, let $k \geq 0$ be the largest integer with

$$m_r := m - kB(2B-1) \;\geq\; 3\Sigma + L, \qquad \text{so} \qquad m_r < 3\Sigma + L + 2B^2.$$

Filler. Let F be the splice of kB copies of K_{2B}. By Lemma 8(1), $m_F = kB(2B-1)$ and $T_F = k(2B-1) \in \mathbb{Z}$. Put

$$\tau' = \tau - T_F = E + m\Big(\rho - \frac{1}{B}\Big) + \frac{m_r}{B}\,.$$

Denominators. By Lemma 9, $D\tau' \in \mathbb{Z}$ and the Q_i are pairwise coprime, so there are integers c_i with $\tau' - \sum_i c_i/Q_i \in \mathbb{Z}$. Let $0 \leq j_i < Q_i$ with $j_i \equiv c_i + 1 \pmod{Q_i}$, and let $Z_i = Z(Q_i, j_i)$. By Lemma 8(2), $T_{Z_i} \equiv (j_i - 1)/Q_i \equiv c_i/Q_i \pmod 1$, $T_{Z_i} \leq Q_i + 2 \leq 3Q_i$ and $m_{Z_i} \leq 3Q_i$. Put $m_g = \sum m_{Z_i} \leq 3\Sigma$ and $T_g = \sum T_{Z_i} \leq 3\Sigma$.

Integer part. Let $l = m_r - m_g \geq L$ and $t = \tau' - T_g \in \mathbb{Z}$. Since $E \asymp x^3$ while $\Sigma, L, m(\rho - 1/B)$ are $O(x^2)$, for s large we get $l \leq 3\Sigma + L + 2B^2 \leq E - O(1) - 3\Sigma \leq t$ and $t \leq E + O(1) + m_r/B \leq 2E + 2 \leq L^2/2 \leq l(l+1)/2$. By Lemma 8(3) there is a rooted tree Θ with l non-root vertices and $T_\Theta = t$.

Conclusion of the construction. Let $W = F \oplus Z_0 \oplus \cdots \oplus Z_4 \oplus \Theta$, rooted at w. Then $m_W = m_F + m_g + l = m$ and $T_W = T_F + T_g + t = \tau$. So G has n vertices, and all ab block vertices are good. This already proves Problem 4 for β. It remains to show that G has no other good vertex.

No other good vertex. The vertex w is a cut vertex, since $m \geq 1$. For a hub h, Lemma 3 and Lemma 6 give $\Delta_G(h) = \delta_h - m\rho_h + T_W > 0$. Let z be a vertex of W that is not a cut vertex of G. It lies in a branch Z of W, one of the copies of K_{2B}, one of the Z_i, or Θ. Let R be the rest of G, so $G = Z \oplus R$ and $T_R \geq \mathrm{Rt}_H(w) \geq a/b$ by Lemma 5. By Lemma 3, $\Delta_G(z) = \delta_Z(z) - m_R\rho_Z(z) + T_R$. If z is pendant, or Z is a clique ($j_i = 0$ included), then $\delta_Z(z) > 0$ and $\rho_Z(z) < 0$ by Lemma 8, so $\Delta_G(z) > 0$. Non-pendant vertices of Θ and the vertex y of Z_i for $j_i \geq 1$ are cut vertices. The remaining case is a clique vertex $z \neq y$ of Z_i, with $j_i \geq 1$; then $\delta_{Z_i}(z) > 0$ and, with $c = 2Q_i$, $j_i \leq Q_i - 1$,

$$\rho_{Z_i}(z) = \frac{2(j_i - 1)}{c(c-1)} \leq \frac{1}{2Q_i}\,, \qquad \text{so} \qquad \Delta_G(z) \geq \frac{a}{b} - \frac{n}{2Q_i} = \frac{a}{b}\left(1 - \frac{b^2 r}{2pQ_i}\right) > 0,$$

because $Q_i \geq x - 1 \to \infty$. Hence $K(G)$ is exactly the set of block vertices, $|K(G)| = ab = \beta n$, and $n \to \infty$ with s. □

Remark 10. **(1)** Without the filler F one takes $m = O(x^2)$ and gets proportions tending to $b/(b+1)$ from below, which already gives Problem 4. The filler only lowers the proportion to the exact value β. Its vertices are nearly neutral, because $\rho \approx 1/B$ and each vertex of K_{2B} has resistance $1/B$ to w.

(2) The choice of x only serves to split the denominators a, $ab+1$, $ab+2$ of Lemma 6 into factors of size $O(x^2) = o(E)$.

(3) The barrier of [2] concerns good vertices lying in a fixed finite family of branches at a cut vertex. Here all good vertices lie in one branch at w, and that branch has unbounded order, so there is no conflict.

(4) The rook graph $K_a \square K_b$ with hubs on the rows, studied in [2], has the same real behaviour. There, though, the vertex-deleted graph has large irreducible denominators such as $a^2b + ab^2 - b - 1$. The clique base removes them: the class of Section 2 is closed under deletion, and its Kirchhoff index depends only on (N, a, a', Y).

5 Explicit graphs

The proof above is not optimised. For $\beta = 2/3$, $b = 3$ it gives $n = 8.4 \cdot 10^7$. Smaller graphs come from the same equation $T_W = m\rho - \delta$ by factoring the denominators into prime powers (with cliques K_q for odd q, where $r = 2/q$). Table 11 lists four of them. A gadget (c, j) is K_c with j leaves on one vertex, and the tree is the tree of Lemma 8(3).

	a, b	gadgets (c, j) at w	tree (l, t)	n	proportion
E1	396, 3	(5,2) (7,6) (11,7) (17,8) (29,18) (41,10)	(32,94)	1772	$297/443 \approx 0.6704$
E2	506, 3	(4,0) (9,2) (5,3) (49,3) (11,9) (19,10) (23,6) (31,23)	(53,133)	2277	2/3
E3	1116, 4	(64,30) (27,6) (5,3) (7,6) (11,3) (19,12) (29,4) (31,12) (47,2)	(62,114)	5952	3/4
E4	705, 4	(4,0) (9,0) (7,2) (13,9) (17,3) (31,25) (47,34)	(15,43)	3735	$188/249 \approx 0.7550$

Table 11 / Explicit graphs $H_{a,b} \oplus W$. In each of them $K(G)$ is exactly the set of block vertices.

To our knowledge E1, E3 and E4 are the first connected graphs other than C_5 with more than two thirds of their vertices good. We make no claim of minimality: nothing has been searched systematically above $n = 10$. In exploration, a variant with W glued at a hub gave $n = 1543$ and proportion $1032/1543$; that graph is not part of the certificates.

6 Limitations

- Problem 3 ($\beta = 1$) is not addressed.
- In Theorem 1, "s large" is not made explicit. This does not affect the theorem. `check_construction.py` carries out the construction with exact arithmetic for 13 values of β from 1/100 to 99/100, and confirms every step on the actual numbers.
- We did not determine the smallest graph with proportion above 2/3.
- The graphs are very dense and have a cut vertex. We do not know a 2-connected family with proportion above 2/3.
- The result has not yet been checked by an independent verifier.

7 Computer verification

All claims are checked with exact rational arithmetic (Python fractions, python-flint) under Fedora 44 in WSL, with Python 3.12, python-flint 0.6, sympy 1.14 and networkx

3. No floating-point value enters a claim. The scripts are in `scripts/` and the logs in `certificates/`.

script	claims (PASS/FAIL, exit code)	time
`check_formulas.py`	Lemmas 3 to 6, 8 and 9: symbolic identities, and brute force on random graphs of the class	1 min
`check_construction.py`	the proof of Theorem 1 on 78 parameter sets, 13 values of β: counts, the equation $\Delta_G(v) = 0$, all inequalities of the last step	1 min
`check_examples.py`	complete $K(G)$ of E1–E4, every vertex type, through Lemma 3	1 min
`check_direct.py small`	$K(G)$ computed vertex by vertex on the actual Laplacian for 8 graphs with n from 24 to 400	3 min
`check_direct.py family`	a graph produced by the proof itself ($\beta = 1/100$, $b = 2$, $s = 2$, $n = 2000$), one vertex per class	5 min
`check_direct.py` with `E1`, `E2`, `E4`, `E3quick`	E1, E2, E4: exact inverse of the actual grounded Laplacian, one vertex per class; E3: block vertex, hub and root	8–90 min

`check_direct.py` uses none of the closed forms of this paper: it builds the adjacency lists and inverts the grounded Laplacian exactly. This is fast because the denominators of these graphs are small; the Kirchhoff index of E2 is 1134673709326253/1314329940.

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